Shaswat Raj

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EDUCATION

Birla Institute of Technology, Mesra — Ranchi, India
Bachelor of Technology (B.Tech) in Computer Science & Engineering — Expected Grad: May 2028

- **Coursework**: Data Structures & Algorithms, Operating Systems, Database Management Systems, System Design, Computer Networks, Object-Oriented Programming, Software Engineering.

TECHNICAL SKILLS

Languages : C++, Python, JavaScript (ES6+), TypeScript, SQL (PostgreSQL, MySQL), HTML5/CSS3
Frameworks & Web : React, Next.js, Node.js, Express.js, Hono, Tailwind CSS, REST APIs, WebSockets
Databases & Cloud : PostgreSQL, MySQL, Redis, Cloudflare Workers, Docker, Git, GitHub Actions, GCP
Core CS & AI : Data Structures & Algorithms, System Design, RAG & Vector Search, Microservices, Web Performance

EXPERIENCE & TECHNICAL LEADERSHIP

Bitotsav '26 (College Cultural Fest) — Nov. 2025 – Present
Engineering Lead & Core Architect — Ranchi, India

- Architected and shipped full-stack event platform serving **5,000+ participants** with zero downtime under peak load.
- Engineered **encrypted dynamic QR code validation**, anti-forgery entry verification, admin panel, and real-time score tracking.
- Designed high-concurrency backend architecture and presented technical specs to college leadership and professors.

Open Source Ecosystem & Developer Tools — 2022 – Present
Maintainer & Core Contributor — Global

- Authored and published **50+ npm packages** accumulating over **1M+ combined monthly downloads**.
- Contributed merged pull requests directly to **Google Gemini CLI** (`google-gemini/gemini-cli`) repository.
- Maintained 40+ high-impact open-source repositories with **15,000+ total GitHub contributions** and **300+ merged PRs**.

FEATURED PROJECTS

Debo – AI Memory OS & RAG Engine | *Next.js, TypeScript, Vector Search, OpenAI, Cloudflare* — 2026

- Developed a personal AI memory operating system transforming notes, voice, and documents into a searchable knowledge graph.
- Engineered Retrieval-Augmented Generation (RAG), vector similarity search, and real-time voice agent context retrieval.

SopKit (30tools) | *Next.js, TypeScript, Cloudflare Workers, Redis, SEO* — 2024 – Present

- Engineered 460+ developer tools, scaling to **4.3M+ monthly requests** and **45K+ active users** with sub-100ms latency.

Telegram Bot Automation Platform | *Node.js, Telegram API, Cloudflare Workers, Redis* — 2024

- Built automated backend microservices, Telegram bots, AI integrations, and media pipelines serving **21K+ monthly active users**.

Kishan AI (AI Agricultural Assistant) | *Python, OpenAI, Computer Vision, Telegram API* — 2025

- Created an AI assistant offering computer-vision crop disease diagnosis, weather telemetry, and multilingual voice support.
- Shortlisted among the **Top 500 teams nationwide** in the prestigious AICTE Atal Puraskar Fellowship (APF) 2025.

HONORS & ACHIEVEMENTS

- **10+ Hackathon Wins**: Placed 1st / Top Tier across 10+ national and regional hackathons for rapid software execution.
- **AICTE APF 2025 Top 500**: Selected in Top 500 teams across India for building AI agricultural solutions.
- **1M+ Monthly npm Downloads**: Authored open-source libraries actively adopted by thousands of developers globally.
- **45K+ Dev.to Followers**: Technical blogger publishing deep dives on web performance, open source, and system design.